PRESENTATION SCRIPT: BLUE IMPULSE SOJOURNER

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Speaker 1:
Hello everyone, we are team Blue Impulse. Today, we'll be presenting our application of the study guide, focusing on the Sojourner Mars Rover.

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Speaker 1:
(Visual intro sequence showing space transit from Earth to Mars)

Today, we take a trip across space to look closely at a ground-breaking milestone in space exploration.

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Speaker 1:
So, what is Sojourner? Over the course of this presentation, we will break down its hardware, its networking capabilities, and its behaviorist AI structure.

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Speaker 1:
Here we see Sojourner, the tiny rover that rolled across Mars. With its small wheels and big courage it gave humanity its first soft glimpse at another world up close.

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Speaker 1:
To give you a sense of scale, Sojourner was NASA's Martian rover that landed on July 4th, 1997. Remarkably, the rover was the size of a microwave!

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Speaker 1:
Now, let's jump into Section 1: Hardware. What makes Sojourner function? We'll pass it over to Speaker 2 to explain.

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Speaker 2:
Thanks, Speaker 1! Computer hardware refers to the physical components of a system including input hardware, processing units, and output hardware.

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Speaker 2:
Firstly, let's look into the Pathfinder Lander.

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Speaker 2:
The Lander's main goal was to observe and image the rover's operation on Mars and transmit data back and forth between Sojourner and Earth. It was essentially a middleman between the rover and Earth

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Speaker 2:
As seen here in the lander diagram, it contained critical components like solar panels, antennas, and meteorology sensors.

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Speaker 2:
Highlighted here is the Imager for Mars Pathfinder, A.K.A IMP, which provided us with

high-resolution visual images that you see in this presentation.

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Speaker 2:

The lander had a flower-like structure with 3 petal shaped structures that contained solar panels and other critical components. Over 2.5 meters of solar cell panels covered the lander's petals to recharge its batteries.

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Speaker 2:
It also contained the Atmospheric Structure Instrument and Meteorology Package (ASI/MET), acting as a Martian weather station tracking pressure, wind, and temperature.

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Speaker 2:
Moving onto the rover itself —the rover contained its own solar panel array , an antenna, a Rocker-Bogie mobility system, a warm electronics box, cameras, lasers, and scientific instruments like the Alpha Proton X-ray Spectrometer.

I will now hand over the topic to Speaker 3 Who will break down how these individual rover systems worked together.

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Speaker 3:
Thank you! Let's look at Sojourner's vision system. Sojourner was equipped with 3 cameras used for navigation, hazard detection, terrain observation, and monitoring its own wheel tracks.

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Speaker 3:
Specifically, it used 1 rear-facing color camera and 2 forward-facing monochrome cameras working alongside 5 laser stripe projectors to create stereoscopic images.

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Speaker 3:
The 5 laser diodes projected light lines onto the ground to detect immediate hazards for reactive navigation, giving real-time local terrain data rather than a pre-mapped layout.

--- [PAGES 32–33] ---

Speaker 3:
On the rear sat the APSX, designed to analyze the elemental chemical composition of Martian rocks and soil.

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Speaker 3:
At the heart of it all was Sojourner's CPU, It used an 8bit Intel 80C85 that ran at a mere 2MHz with 0.56MB ram. That is roughly 0.05% the clock speed of a modern 4GHz Cpu that our modern computers have. This processor executed instructions and processed data collected from all these sensors with less ram than our modern calculators!

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Speaker 3:
So for movement on the Martian Terrain, it used 6 aluminum wheels connected to a

Rocker-Bogie suspension system, allowing it to navigate rocky obstacles far better than a

standard 4-wheel setup would allow.

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Speaker 3:
Finally, an antenna mounted on top handled communication.

Speaker 4 will explain how that network worked!

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Speaker 4:
Thanks! Let's examine Section 2: Networks, and how Sojourner communicated back to Earth.

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Speaker 4:
What is a network?, by definition, it links independent systems together to share data and resources.

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Speaker 4:
The communication path followed a three-way relay: Earth communicated with the Lander, and the Lander communicated directly with Sojourner.

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Speaker 4:
For command flow from Earth to Sojourner: first, engineers prepared command sequences on Earth. These were transmitted over deep-space links to the Lander, which stored them.

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Speaker 4:
Sojourner would then establish a radio session with the lander, retrieve the commands, execute them, and engage in local autonomous navigation when driving.

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Speaker 4:
For telemetry flow back to Earth: Sojourner collected sensor readings, operational measurements, and scientific results, organizing and storing them in it's memory.

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Speaker 4:
It transmitted these to the lander, which buffered the data and relayed it to Earth for team examination and future planning.

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Speaker 4:
The link between the rover and lander used Ultra High Frequency (UHF) radio at ~459 MHz, transferring data at roughly 9600 bits per second in a half-duplex configuration.

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Speaker 4:
Half-duplex means data flows both ways, but not at the exact same time—transmission and reception occur in separate time windows.

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Speaker 4:
To prevent lost rovers, Sojourner ran periodic heartbeat checks. If communication dropped during a drive, it would pause, attempt recovery, or retreat back toward the lander to restore contact.

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Speaker 4:
These network setups were crucial because radio signals between Earth and Mars took 10 to 15 minutes each way, and Sojourner's small RAM couldn't store massive maps. It had to make decisions on its own.

Speaker 5 will explain how sojourner's behaviorist AI funtioned.

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Speaker 5:
Thank you! Section 3 covers Behaviorist AI—how Sojourner decides what to do.

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Speaker 5:
Behaviorism in AI focuses on observable actions driven by environmental stimuli, hinging on two main concepts: situatedness and embodiment.

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Speaker 5:
There's an interesting debate on whether Sojourner counts as true AI. While it doesn't use machine learning, it is a behavior-based robot built on stimulus-response loops.

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Speaker 5:
Here is how its stimulus-response loop works: the 5 laser stripers project light on the ground while the camera looks for line deformations caused by rocks or drop-offs. Low-level safety behaviors instantly override high-level goal seeking if a threat appears.

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Speaker 5:
If a bumper hits a rock or a laser sees a hole, the rover automatically stops, backs up, and turns around. It repeats this loop until it reaches its target coordinate.

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Speaker 5:
Why did we choose to study Sojourner? Because it is a pure example of reactive behaviorism. Later rovers like Curiosity or Perseverance rely much more on heavy algorithmic, rule-based autonomy while sojourner had a simpler reactive AI that was substantially easier to study and teach.

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Speaker 5:
In conclusion, Sojourner functioned as one unified system: Hardware sensed physical actions, Networks handled communication, and Behaviorist AI managed decision-making.

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Speaker 5:
Thank you for listening!
